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Investigating the Validity of Predicting Atlantic Hurricane Strength with Ocean Surface Temperature

Introduction and Research Question

Hurricanes, also called “typhoons” or “cyclones” based on geography, are defined by the National Oceanic and Atmospheric Administration (NOAA, 2020) as “severe storms with high-velocity winds that rotate around a central, low-pressure core”. Several factors contribute to hurricane formation, with one primary factor being the presence of warm ocean surface waters above 27°C, or 80°F, that serve as an energy source for the hurricane. Warm ocean water facilitates evaporation, adding moisture to the air and causing latent heat release when that moisture condenses, which powers storm systems and boosts storm size and intensity (NOAA, 2020).

In the United States, hurricanes most frequently strike states along the Atlantic coast or the Gulf of Mexico, both part of a geographic region known as the Atlantic Main Development Region (MDR). The Atlantic MDR is a region covering the tropical waters from the west coast of Africa to the Gulf of Mexico

The research question I am answering is: *Have hurricanes developing in the Atlantic Main Development Region been increasing in intensity over the last 50 years (1975-2024)? How does ocean warming due to climate change and variations in water temperature over time affect the strength of developing hurricanes?*

Studies have shown that increases in ocean temperatures due to human-caused global warming are likely linked to global increases in hurricane strength (Masters, 2026). Increased hurricane wind speeds lead to greater property damage, environmental destruction, and potentially loss of life. Given the high destructive potential of hurricanes, it is important to understand how environmental conditions are linked to hurricane formation, as well as how global warming could be worsening hurricane strength. Understanding this correlation is valuable for arguing for stronger environmental policy to limit human activities that contribute to ocean warming in the future.

Data

I used two main datasets for this project. For ocean surface temperature, I used the NOAA’s “Climate at a Glance” Global Time Series tool, which calculates global temperature

departure using NOAA's GlobalTemp dataset. NOAA GlobalTemp combines sea surface temperature data and land surface temperature data from several NOAA datasets to create a complete representation of global temperature trends over time (Huang et al.).

Using the online tool, I chose to see monthly average temperature departure data for land and ocean in the Atlantic MDR (Main Development Region) from 1975 to 2024, with the base period being 1910-2000. The Atlantic MDR is a region covering the tropical waters from the west coast of Africa to the Gulf of Mexico. The tool does not offer solely ocean temperature data, but because the Atlantic MDR region primarily covers water, I feel comfortable using the land and ocean data together, as I believe the land data will not skew the dataset significantly as a whole. NOAA GlobalTemp covers the years 1850-2025 for all regions globally. I chose to only focus on the years 1975 to 2024, as this is the time period that matches up with my hurricane dataset, which I discuss below. The main variables for this dataset are Date (year, month) and Departure from Average in °C. A temperature departure is a measure of how much the observed average temperature for a given month deviates from the expected temperature based on a long-term average (NOAA 2026). This allows us to see changes in global temperature over time compared to a reference value. No data cleaning was required for this dataset, as I just picked the relevant parameters using the online tool and generated a .csv file with data from the region and time period that I wanted.

The second dataset I used was the National Hurricane Center's Atlantic Hurricane Database (HURDAT2), a dataset generated through post-storm analysis of all known hurricanes originating in the Atlantic Basin. I used a cleaned version created by a user on Kaggle, who added variable labels for each column, as well as identified each storm using its name rather than its ID number. This makes the data much easier to work with. HURDAT2's main variables include Date (year, month, day), Location (latitude & longitude), Status, Category, and Max. sustained speed in knots. The original HURDAT2 dataset contains hurricane records from 1851-2025, but the cleaned version only contains 1975-2024, which is the temporal range I will be using in my analysis. I still had to do some processing on this dataset by organizing the data by year and month rather than by storm name, as well as finding the average wind speed of all hurricanes during each month, as my analysis focused primarily on how monthly ocean temperature data correlates with hurricane strength. Therefore, it makes the most sense to average out the windspeed of all the storms observed in a certain month.

Methods

All graphs were generated using the Excel software. The T-test was conducted using Excel's built-in Data Analysis tool.

Time series

I used time series analysis to observe how the average strength (windspeed) of hurricanes formed in the Atlantic MDR changed over time. To do this, I first calculated the average wind speed of all the storms that had formed in the region during a given month of a given year. Then, I plotted the average wind speed on the x-axis and the year on the y-axis of a line graph in order to visualize the change over time. A time series plot is effective because it allows us to see any long-term trends and identify seasonal cycles.

Time Series Decomposition

Using the time series I computed previously, I conducted a decomposition to understand the overall trend and seasonality of changes in hurricane windspeed over time. This is an appropriate method to understand our hurricane data better, and isolate the factors contributing to variations. I used the additive decomposition model as I predict the seasonal fluctuations to be consistent over time. I split the above time series into three components: the trend, the seasonality, and the residuals, and plotted each on its own time series plot for visualization and analysis.

Regression analysis

I conducted a simple linear regression analysis on my data to determine whether there is a substantial linear relationship between the monthly average temperature departure and the average windspeed of the storms originating during that month. This technique is appropriate because it allows me to visualize the data and test whether variation in windspeed can be predicted by variations in temperature departure, which is indicative of whether warmer-than-average waters lead to more severe storms.

The response variable is the average windspeed (knots) and the predictor variable is average monthly temperature departure (°C).

Null hypothesis: Average windspeed is not linearly associated with average monthly temperature departure.

Alternate hypothesis: Average windspeed is linearly associated with average monthly temperature departure.

T-test

I conducted a two-sample t-test on my data to determine if there is a significant difference between the hurricane windspeeds observed during comparatively low ocean temperatures and the windspeeds observed during comparatively high ocean temperatures. In order to do this, I separated my windspeed data into two groups based on the ocean temperature departure during a given month. Our ocean temperature dataset reports temperature based on departure, not just the raw values, making it slightly tricky to determine what should be considered “low” and “high”

temperatures. I decided to split the data in half, with any temperature departure value below 0.5 °C designated as “low temp”, and any value above 0.5 °C designated as “high temp”. This is because temperature departure measures how much the observed temperature for a given month deviates from the expected temperature based on a long-term average, so a low departure value indicates lower than expected or slightly above expected temperatures, and a high departure value indicates higher than expected temperatures.

Before doing the t-test, I checked the variances of my two groups and found them to be roughly equal. Therefore, I conducted a two-sample t-test assuming equal variances. I used the Data Analysis tool in Excel to conduct the test using my organized data.

Null hypothesis: there is no difference in means between the average windspeed of hurricanes formed in low-temperature waters and the average windspeed of hurricanes formed in high-temperature waters.

Alternative hypothesis: there is a difference in means between the average windspeed of hurricanes formed in low-temperature waters and the average windspeed of hurricanes formed in high-temperature waters.

Chi-Squared Test of Independence

I conducted a Chi-squared test of independence to determine whether monthly average storm intensity (based on windspeed) and ocean temperature are associated with each other, which would suggest that higher water temperatures cause more intense hurricanes. To do this, I first had to organize these variables into categories, because the original data is numerical rather than categorical. For ocean temperature, I used the categories of “low temp” and “high temp” that I created for the t-test I conducted previously. For the storm intensity, I created three categories, “low intensity”, “medium intensity”, and “high intensity”. “Low intensity” months are months where the average storm windspeed is below 27 knots. “Medium intensity” months are months where the windspeed is between 27 and 54 knots. “High intensity” months are months where the windspeed is above 54 knots. I picked these categories to create evenly spaced categories within the range of our dataset.

To calculate the Chi-squared value, I created a contingency table that showed observed counts, and a table that showed expected counts if the null hypothesis were true. Then, I calculated the Chi-square statistic and the degrees of freedom, and compared it to a p-value lookup table to find the p-value of the test. All the relevant tables are in the “Results” section below.

Null hypothesis: there is no association between monthly average storm intensity and monthly average ocean temperatures.

Alternative hypothesis: there is an association between monthly average storm intensity and monthly average ocean temperatures.

Results

Time Series

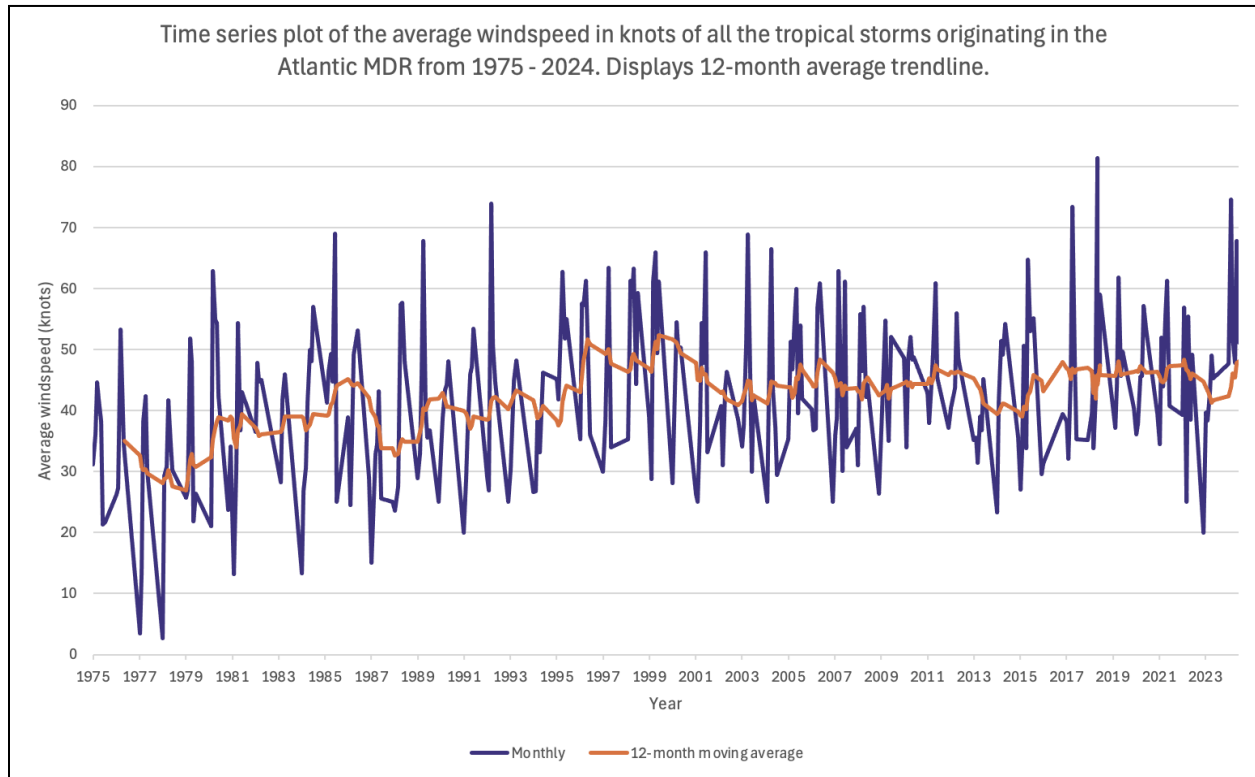
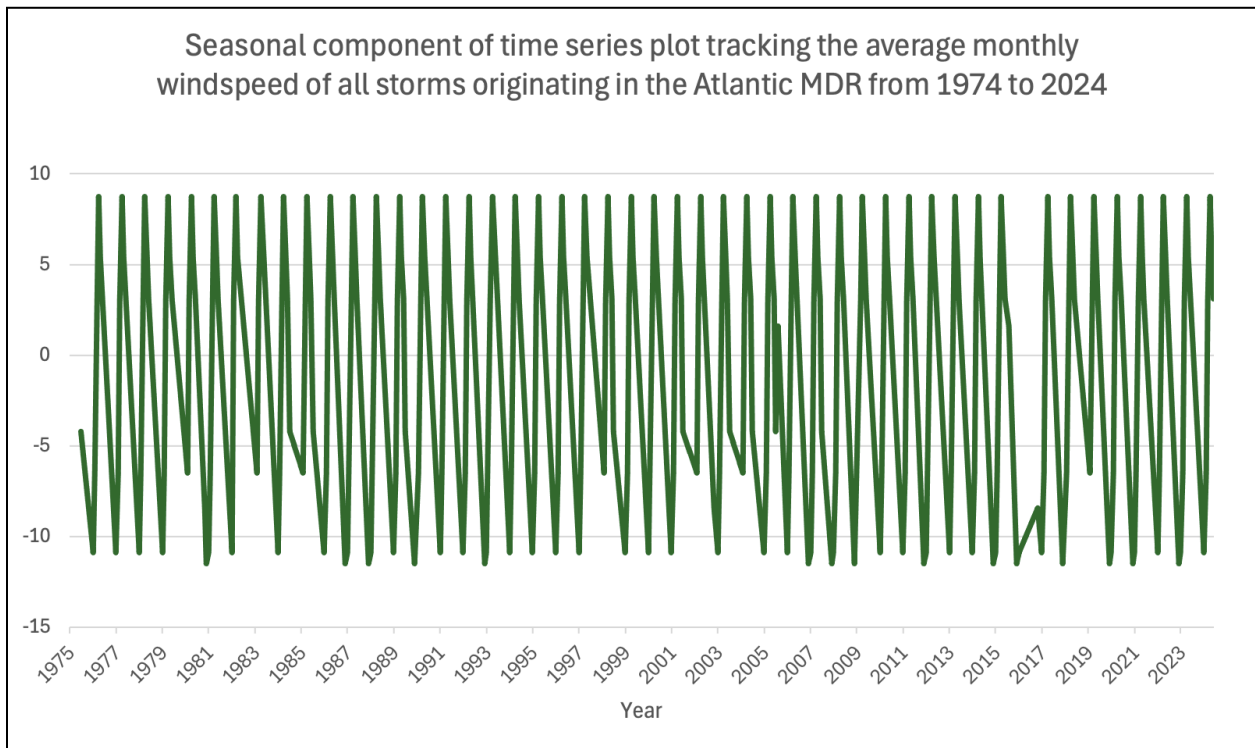
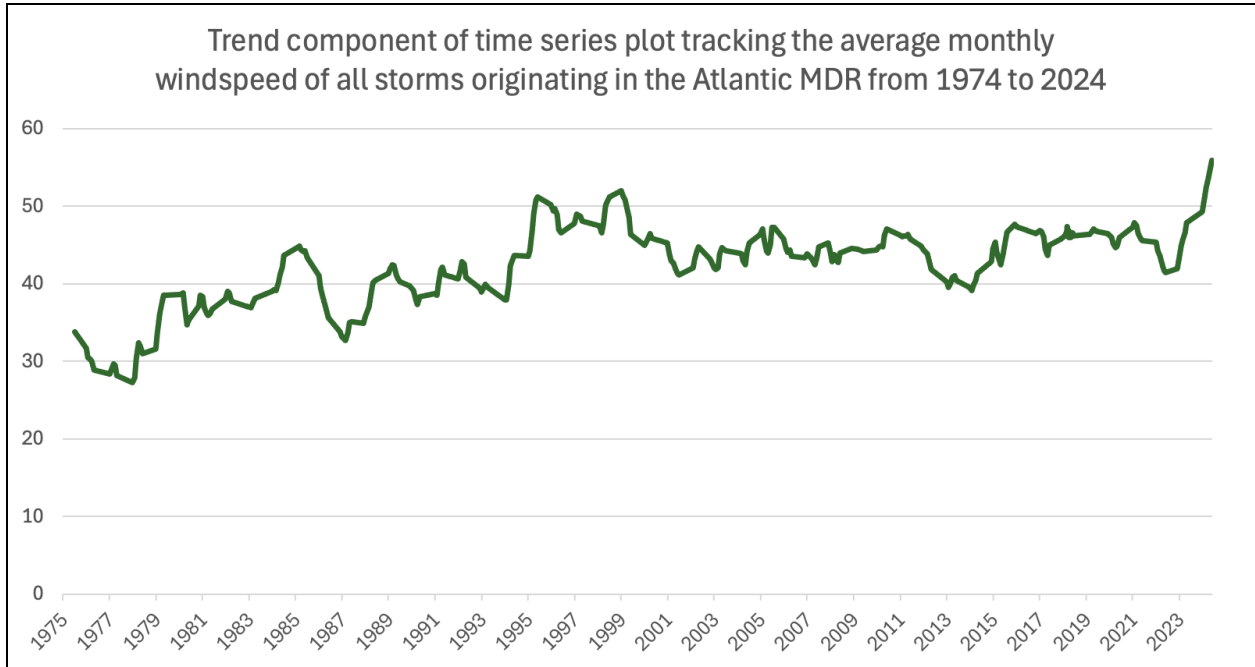
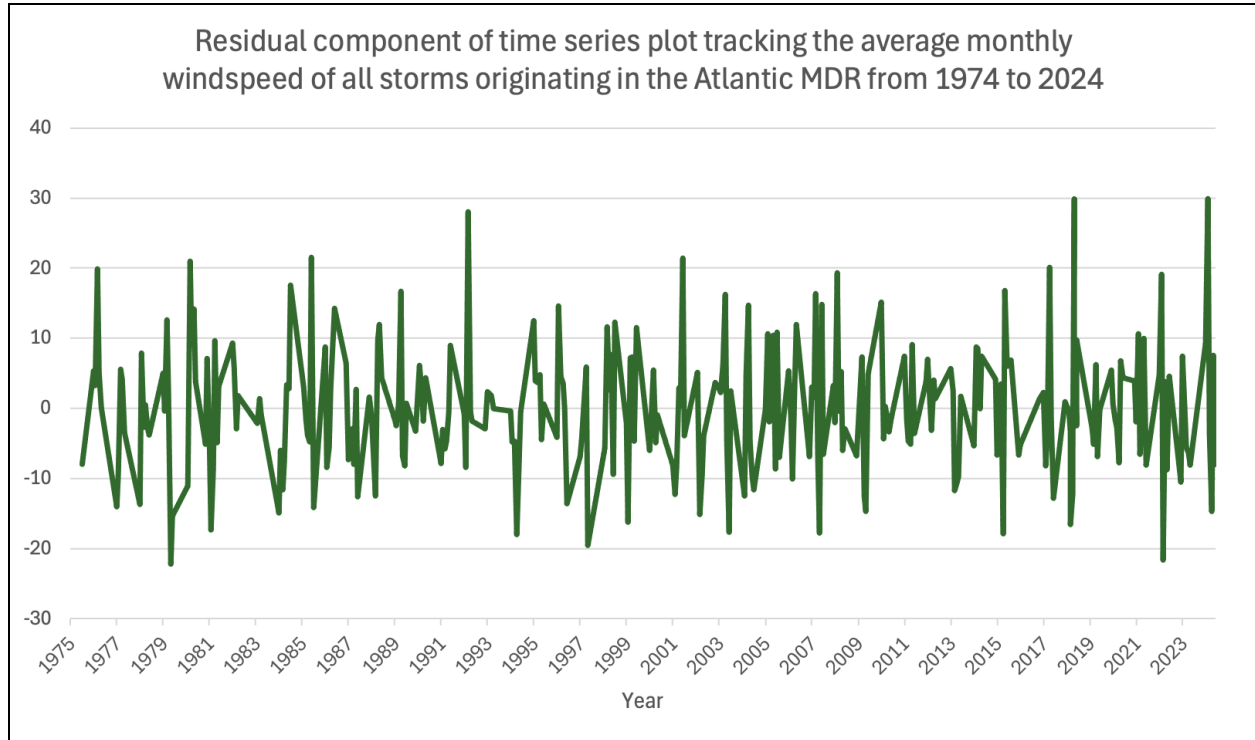


Figure 1. Time series plot of the average windspeed (knots) of all the tropical storms originating in the Atlantic MDR between 1975-2024. The 12-month trailing moving average is displayed on the graph in orange. Only months with recorded hurricanes are plotted - primarily the Atlantic hurricane season of June through November.

The time series plot shows a very slight upward trend in average storm windspeed from 1975 to 2024, but the trend is not clear or significant, even when using a 12-month moving average to smooth the data. This suggests that there has not been a large increase in the average storm strength since 1975. The seasonal component appears to be strong, with large fluctuations in average windspeed occurring from month to month within a given year. However, it is difficult to tell on this graph during which specific months the windspeed will peak versus dip.

Time Series Decomposition





Figures 2, 3, 4. Plots of decomposed components of Figure 1, the time series plot of the average windspeed (knots) of all the tropical storms originating in the Atlantic MDR between 1975-2024. The additive decomposition model was used. The trend, seasonal, and residual components are separated into three different time series plots.

This time series decomposition produced some interesting results regarding our hurricane data. The trend component appears to be increasing from 1975 to present, although it is still not very smooth even after using a 12-month moving average to smooth it. It shows an overall increase in the average windspeed of storms in the Atlantic MDR, even after having the seasonal and residual components removed. The seasonal component is quite strong, contributing up to 10 knots of additional windspeed to storms during hurricane season. The residual component is varied and does not show a clear trend in any direction.

Our decomposition shows us that there is an increasing trend towards higher-windspeed hurricanes from 1975 to 2024, which suggests that hurricanes in the Atlantic MDR have been increasing in intensity over time.

Regression Analysis

The linear regression model generated the plot below:

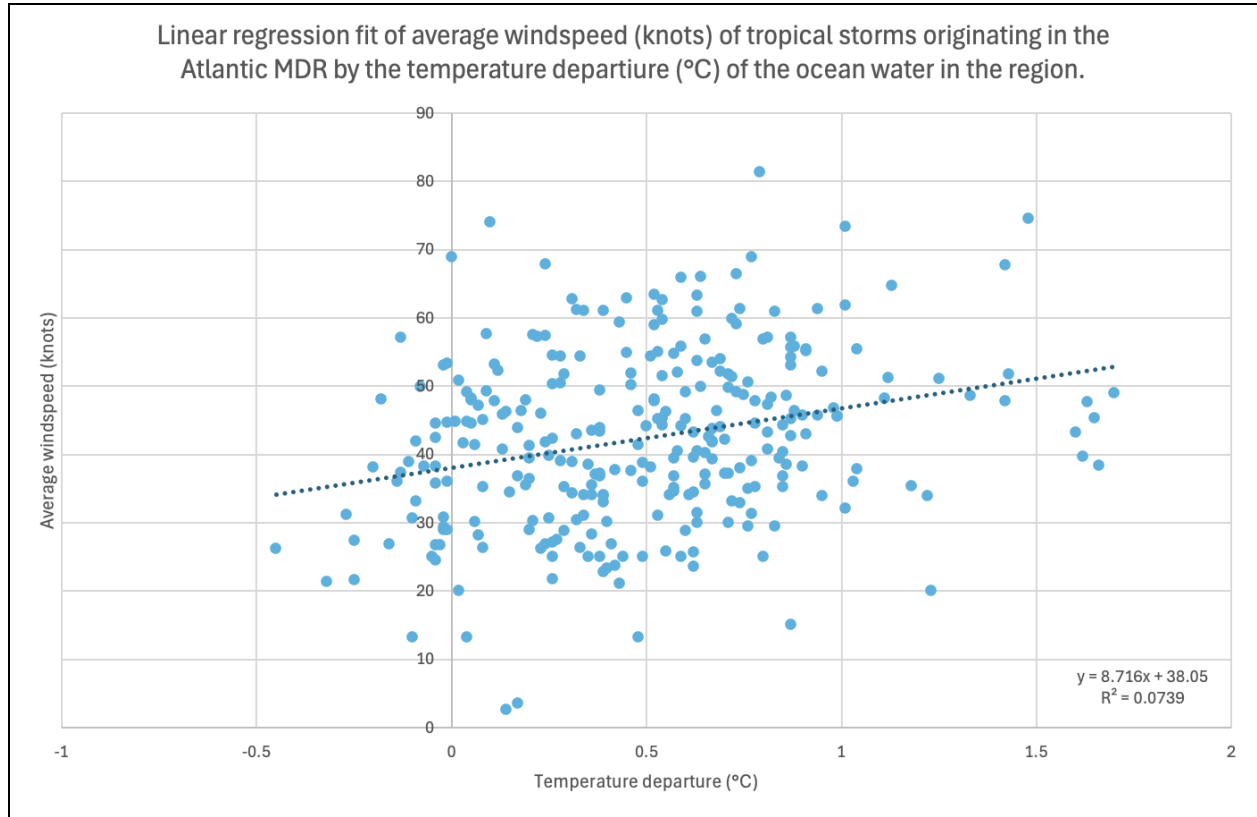
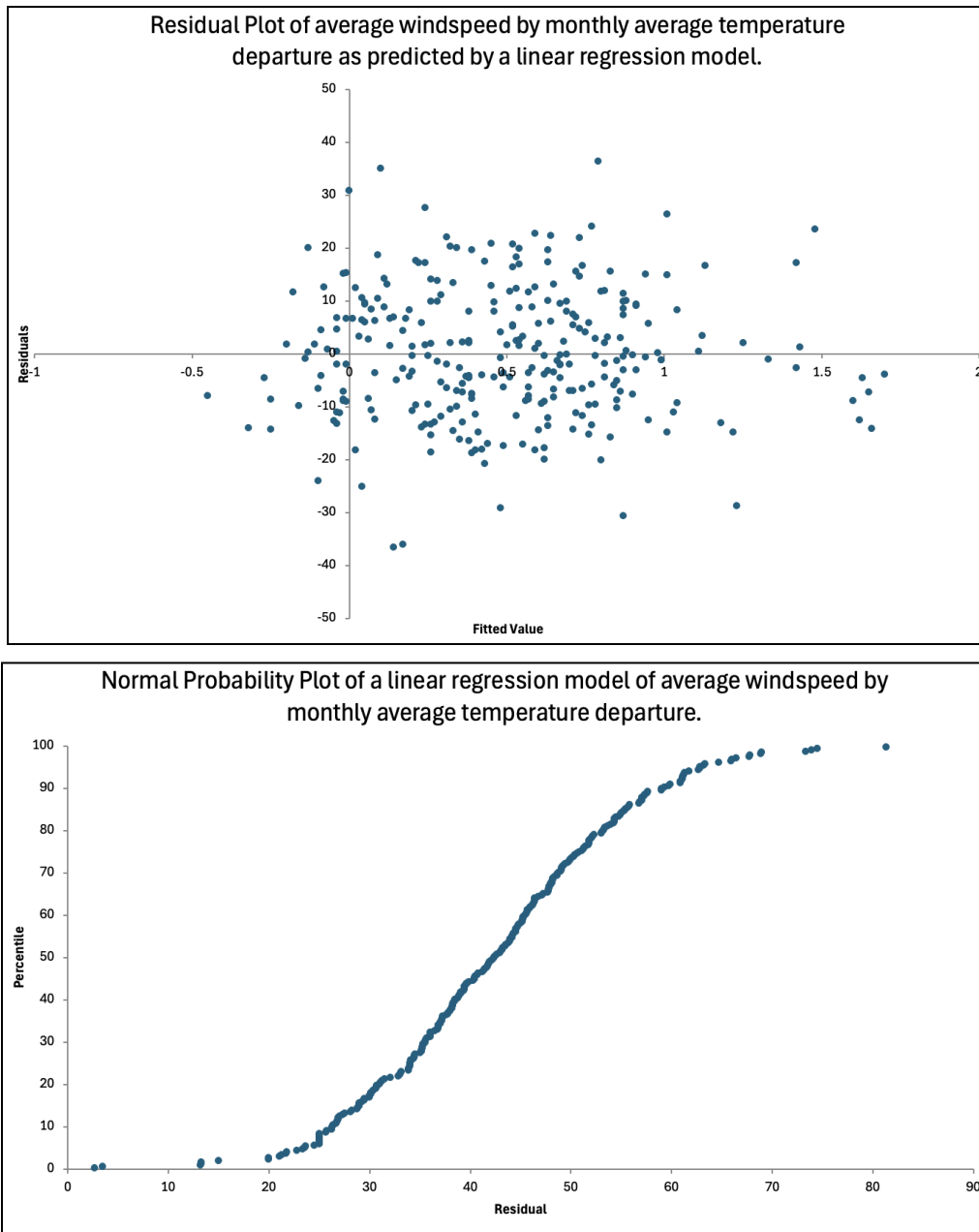


Figure 5. A simple linear regression plot with average windspeed (knots) of tropical storms formed in the Atlantic MDR on the y-axis as the response variable and average monthly temperature departure (°C) of the ocean water on the x-axis as the predictor variable. The regression equation and R-value are shown on the plot.

The R^2 -value for this regression is 0.0739. This means that only 7.39% of the variation in a storm's average windspeed can be explained by variations in the temperature departure of the water over which the storm formed. This is quite a low percentage, suggesting that the majority of variation in windspeed is not due to the temperature departure of the ocean water. I also generated a residual plot as well as a Q-Q (normal probability) plot below:



Figures 6 and 7. Residual plot and normal probability plot for the linear regression generated in Figure 5.

Figure 6 is a residual plot, which shows the residuals of the regression plotted as a scatter. The scatter appears to be random around 0 and does not show a clear curve or funnel shape, which could indicate nonlinearity. Figure 7 is a Q-Q or Normal probability plot, which checks if residuals are normally distributed. This Q-Q plot has heavy tails that curve and create an S-shape, indicating potential non-normal distribution of residuals.

Two-sample t-test

Table 1. Results summary table for a two-sample t-test assuming equal variances. Table includes means and variance for both groups, number of observations, degrees of freedom, t-statistic, and one and two-tailed p-values.

t-Test: Two-Sample Assuming Equal Variances

	Windspeed, low temp	Windspeed, high temp
Mean	38.73991424	45.9503746
Variance	162.6870052	134.553041
Observations	146	147
Pooled Variance	148.571683	
Hypothesized Mean Difference	0	
df	291	
t Stat	-5.062863941	
P(T<=t) one-tail	3.6695E-07	
t Critical one-tail	1.650106758	
P(T<=t) two-tail	7.339E-07	
t Critical two-tail	1.968149554	

The t-Statistic of this t-test negative, indicating that the mean windspeed of hurricanes formed in low-temperature waters is higher than the mean windspeed of hurricanes formed in high-temperature waters.

Both the one-tail and two-tail p-values are below 0.05. In fact, they are even below 0.001, which indicates a strong significance for this test. We can therefore reject the null hypothesis that there is no difference in means between the average windspeed of hurricanes formed in low-temperature waters and the average windspeed of hurricanes formed in high-temperature waters.

Chi-Squared test of Independence

Table 2. Contingency table displaying the distribution of observed values of average monthly hurricane strength and ocean water temperatures. Sorted into categories by hand based on the original numerical dataset.

OBSERVED	Low intensity		Medium intensity	High intensity	TOTAL
Low temp	29		99	18	146
High temp	7		104	36	147
TOTAL	36		203	54	293

Table 3. Contingency table displaying the distribution of expected values of average monthly hurricane strength and ocean water temperatures. Sorted into categories by hand based on the original numerical dataset.

EXPECTED	Low intensity	Medium intensity	High intensity	TOTAL
Low temp	17.9	101.2	26.9	146
High temp	18.1	101.8	27.1	147
TOTAL	36	203	54	293

Table 4. Results summary table for a Chi-Squared test of independence for the variables above. Table includes Chi-squared value, degrees of freedom, and p-value found from a Chi-square distribution lookup table.

X²	19.65
df	2
p-value	<0.005

The Chi-squared test for independence returned a X^2 value of 19.65. At a df value of 2, the p-value for this test is less than 0.005. The lowest p-value displayed on the lookup table I used was 0.005, and my Chi-squared value was even lower, so I cannot calculate the exact p-value. However, this is a p-value lower than a significance level of 0.05, meaning that we can reject the null hypothesis that there is no association between monthly average storm intensity and monthly average ocean temperatures. Our Chi-squared test suggests that hurricane windspeed and the temperature of the water over which they formed are associated variables.

Discussion

Overall, our findings suggest that hurricanes originating in the Atlantic Main Development Region, including parts of the Atlantic Ocean and the Gulf of Mexico, have been getting stronger over time, and it is likely that an increase in Atlantic MDR ocean temperatures have contributed to this increase in hurricane severity.

Time series analysis allowed us to see that even with seasonal and residual components removed, there is an upward trend in the average windspeed of hurricanes developing in the Atlantic MDR, made clearer by the decomposition plots. The trend showed an increase of roughly 20 knots of windspeed from 1975 to present, which is a substantial amount of speed.

The linear regression plot showed weak evidence of a linear relationship between ocean temperatures and storm windspeed, with only 7.39% of the variation in windspeed being explained by variations in the temperature departure of the water over which the storm formed. However, a linear relationship is just one of many possible relationships that two variables can have, and just because our data does not fit a linear regression, does not mean that there is no relationship between water temperature and windspeed - it merely suggests that these variables do not have a linear relationship.

The t-test and Chi-squared tests both show evidence of a relationship between monthly average ocean temperature and the average windspeed of hurricanes formed during that month. The t-test strongly suggests that hurricanes formed in warmer waters have a higher mean windspeed than hurricanes formed in cooler waters, while the Chi-squared test corroborates this by suggesting that these two variables are in fact associated. Both the t-test and Chi-squared test produced very small p-values, indicating that the observed results are very unlikely to have happened due to random change.

All of these tests were effective at helping to answer our research question by demonstrating that hurricanes have been increasing in intensity over the last 50 years, and that increasing water temperatures likely have an effect on this increasing intensity. Even if the test did not provide a “successful” outcome, like the linear regression plot, it gave insight onto what other potential relationships these two variables could have, making it helpful in its own way.

Some assumptions had to be made due to the scope and time frame of this project. Firstly, I do not know how to code in R, therefore all the analysis was done using Excel. This was alright

most of the time, but the time series was tricky, as the seasonal periods were irregular. The HURDAT2 dataset tracks all known hurricanes in the Atlantic MDR, but during months where there are no hurricanes (primarily the winter and spring months), there is no monthly data. Therefore, the seasonal periods were irregular across years, as some years saw hurricanes during 7 months, but some only saw hurricanes during 4 months. Time series analysis of irregular seasonal periods should be done using more advanced analysis techniques like MSTL (Multiple Seasonal-Trend decomposition using LOESS), which is accessible through coding languages like R. However, because I do not use R, I decomposed the time series by hand using classical decomposition, which could have led to inaccurate results due to the irregular nature of my data.

Additionally, the choice to measure hurricane intensity using a monthly average of all the hurricane windspeed data during a certain month was made to simplify the project on my end. However, it likely created unreliable data, as in the HURDAT2 dataset, most storms are tracked throughout its path, and its windspeed was recorded multiple times, meaning that not all the data is fully independent from each other, and some storms may be over/underrepresented due to how many times they are tracked throughout their paths. It is a large assumption to make that the monthly average hurricane windspeed is a good predictor of overall hurricane intensity over time, and this assumption could result in incorrect results.

If given more time, I would find a better way to quantify “hurricane intensity”, perhaps using maximum windspeed instead and doing more detailed analysis that separates the different individual storms occurring in a given month. I would also attempt to create maps using GIS software to visualize how ocean water temperatures vary across the Atlantic MDR, as the region is quite large and it is a large generalization to use only one monthly mean water temperature for the whole region. I wanted to attempt making maps, but unfortunately ran out of time to complete it.

Conclusion

Using the results from our analysis, there is evidence to suggest that hurricanes developing in the Atlantic Main Development Region have been increasing in intensity over the last 50 years (1975 to 2024). There is also evidence to suggest that ocean warming has played a role in the increase in hurricane strength, as warmer ocean waters have been shown to lead to hurricanes with faster windspeeds that are able to do more destructive damage where they make landfall. The oceans have been gradually warming over time, and this environmental issue has come to the public attention more in the past few decades, as we continue to learn more about the ramifications of climate change, and in this case, specifically how climate change worsens natural disasters. Using the analysis completed here today, it is undeniable that the continued ocean warming will lead to more severe extreme weather events that have the potential to destroy property, damage ecosystems, and even take lives. Therefore, it is imperative that steps be taken to slow down the continued warming of the world’s oceans.

Generative AI Acknowledgement

No generative AI was used in the process of this project.

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